

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES / NO

YES

If yes, please indicate which ministries and/or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

The Ministry of University and Research (MUR) plays a central role in promoting open science and data access. The 2021 recommendations have been integrated into national research policies, particularly in the National Research Program (PNR) and the National Open Science Plan (PNSA). The 2021 recommendations have been widely promoted in events organized by academic and research institution stakeholders (e.g., CoPER Open Science Working Group, AISA) and also by other types of institutions (e.g., Museo Galileo).

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES / NO

NO

The Italian version of the 2021 Recommendation has not been produced; However, several communication and dissemination articles in Italian, which mention these recommendations, have been published over the last few years.

If yes, which language(s)?

1.3 Have there been awareness raising activities on open science, including all its key elements associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.q, 16.h)

YES / NO

If yes, please provide details on the activities organized or foreseen and links as relevant.

YES

Several awareness events related to open science have been organised at the national and local levels.

During the G7 Science and Technology Ministers' Meeting held in Italy from July 9 to 11, 2024, the ministers underscored the pivotal role of open science in fostering innovation and addressing global challenges. Key conclusions regarding open science included:

1. **Promotion of Open Science Practices:** The ministers advocated for the equitable and responsible dissemination of scientific knowledge and research outputs. They emphasized the importance of making publicly funded scholarly publications and scientific data accessible to all, thereby enhancing the productivity and quality of research outputs that are Findable, Accessible, Interoperable, and Reusable (FAIR).
2. **Enhancement of Research Security and Integrity:** While promoting openness, the ministers also highlighted the necessity of implementing effective risk mitigation measures to protect against foreign interference in research and innovation.
3. **Improvement of Science Communication:** Recognizing the importance of responsible science communication, the ministers aimed to enhance public trust and inform evidence-based policies.

Most of the dissemination events have addressed specific activities related to implementing the open science approach, such as the change in the scholarly communication model, the need to reform the copyrights model and the introduction of appropriate practices for data management.

Participation and interest in these activities have grown significantly in recent years, driven in part by mandates introduced by funding agencies and research organizations. For instance, all research projects funded through the Next Generation Europe initiative must incorporate elements that contribute to the transition toward open science.

At the national level, awareness has also been boosted by the strong participation of Italian Research Performing Organizations (RPOs) and universities in international initiatives focused on open science implementation. Notable examples include CoARA [53 Italian members], the EOSC Association [24 Italian members], the Network of Open Science Competence Centers [22 Italian members], and copyright-related initiatives. Through their involvement in these initiatives, members have actively discussed and promoted best practices at the local level, amplifying awareness and facilitating the removal of barriers to open science adoption.

Awareness-raising efforts have also expanded significantly within individual institutions. Many universities and RPOs have established specialized Open Science units, including dedicated task forces for training and outreach (e.g. University of Milan, University of Bologna, CNR). Additionally, new university courses and dedicated lecture series within existing programs (e.g., Scuola Normale) (see Section 4 of this questionnaire) have been introduced to educate students on open science principles.

Relevant information about OS activities is disseminated through the open.science.it portal. It is the Italian reference point for Open Science and Open Access to research results. It provides resources and information on various related aspects, such as research data management, Open Science policies, and national and international initiatives in this field.

The website is structured to facilitate access for different user groups, including researchers, institutions, data stewards, funding bodies, and citizens, offering tailored content for each category.

Additionally, the portal features updated news, events, and in-depth articles to promote the dissemination of Open Science practices in Italy.

The Open Science Cafè [<https://www.icdi.it/it/attivita/competence-centre/open-science-cafe>] represents another initiative that has largely contributed to raising awareness. It consists of a series of informal events designed to engage the Italian research community in discussions about various Open Science topics. Hosted by the ICDI Competence Centre, these cafés aim to inform and foster dialogue on data sharing, research transparency, and open access publishing. Each session typically lasts one hour,

focusing on a specific theme and encouraging active participation from attendees.

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science, in publicly funded research?

YES / NO

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant

YES

The operational work plan for implementing the National Open Science Plan, prepared in 2024, will be refined and completed in 2025 to include actions extending through 2027. In the meantime, the process of implementing the national infrastructure for Open Science outlined in this plan will continue to progress. Following a highly competitive selection process, this infrastructure has been chosen as one of the nodes to be included in the first implementation phase of the EOSC federation. By the end of 2025, it is expected that this node will define governance and technical solutions to ensure interoperability with other nodes in the federation. In addition, the Italian Ministry of Research is considering supporting and funding Open Research Europe (ORE). This collective, non-profit service will be supported for an initial period of five years starting in 2026.

Furthermore, the Italian Ministry of University and Research (MUR), jointly with other European funding and research-performing organizations, has committed to concretely contribute to the process of evolving the open-access publishing platform Open Research Europe (ORE) into a collectively funded platform while ensuring its equity, transparency and sustainability. The final aim is to create a pan-European partnership to sustain ORE, advance open access publishing, and foster high-quality research. The new platform is set to operate for an initial period of five years, beginning in 2026.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy, strategy or plan of action on science, technology and innovation (STI)?

YES/NO

YES

The National Research Programme (PNR), established under Legislative Decree 204/1998, is the foundational document that guides Italy's research policies, with its implementation coordinated by the Ministry of University and Research in collaboration with State administrations.

On the European and international fronts, the PNR synthesizes the various intervention lines pursued by central and regional public administrations within the national research and innovation system. It effectively integrates diverse initiatives, contributions, and regional specificities, in full compliance with the distribution of regulatory and administrative competencies.

The PNR aims to foster greater harmony and enhance the coordination of research policies at European, national, and regional levels, thereby strengthening the presence and competitiveness of Italian researchers both within the European Research Area and on the global stage.

The most recent iteration of this plan, covering the period 2021–2027, was approved in December 2020. It is the product of extensive discussions initiated by the Ministry of University and Research with the scientific community and State and regional administrations—and, for the first time, extended through public consultation to stakeholders from both the public and private sectors and civil society.

Designed to support the United Nations Sustainable Development Goals (SDGs), the priorities of the European Commission, the objectives of the 2021–2027 Cohesion Policy, and the Next Generation EU initiative, the plan represents a comprehensive strategy for advancing Italy's research landscape.

Approximately 250 leading experts from the national scientific community contributed to drafting the document, working alongside the administrative structure of the Directorate-General for the Coordination and Valorization of Research and its Results.

The National Plan for Open Science (PNSA) and the National Plan for Research Infrastructures, are both included as key horizontal components of the 2021–2027 PNR.

PNSA encompasses all the listed open science elements.

2.2 In your country, is there a policy, or a set of policies and/or legal framework at national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science?

YES / PARTLY /

If yes or partly, please attach/upload machine-readable **PDF file(s)** of the policy(ies) and provide the following information for each: Title of the policy or legal instrument, the authority in charge of the development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

PARTLY

The legal framework at the National level for publications [art 4, par 2, 3, 4 law 7 October 2013, n. 112] regulates open access to them. The established mandate is lighter than the one of the European Commission in its funding programmes.

Concerning research data, Italy has implemented the 2019 EU Directive on Open Data [<https://eur-lex.europa.eu/legal-content/IT/TXT/PDF/?uri=CELEX:32019L1024>].

Currently, in Italy, the rules on Open Data are regulated by the recent version of Legislative Decree 36/2006, which has been adapted to essentially reflect what is established by the Open Data Directive. The Agency for Digital Italy (AgID) has published specific guidelines to clarify the technical and implementation aspects of the new regulation. Research institutions are implementing these guidelines at different speeds. A similar implementation process is ongoing in organizations involved in the building and maintenance of EU research infrastructure nodes.

2.3 In your country, are there specific policy instruments that aim to promote open science?

YES/ UNDER DEVELOPMENT/NO

If yes or under development, please provide details and links as relevant, including for each instrument: Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

Yes, in Italy, there are several **policy instruments** aimed at promoting **open science**, aligning with European Union directives and fostering transparency, accessibility, and collaboration in research. Some key instruments include:

1. National Open Science Plan (Piano Nazionale per la Scienza Aperta) - 2023

- Approved by the **Ministry of University and Research (MUR)** in 2023, this plan aligns with **EU Open Science policies** and the European Open Science Cloud (EOSC).

- It establishes guidelines for **open access publishing, research data sharing**, and **FAIR principles** (Findable, Accessible, Interoperable, Reusable).
2. National Research Plan (Piano Nazionale della Ricerca - PNR)
- The **PNR 2021-2027** includes open science as a strategic objective, promoting **open access** to scientific publications and research data.
 - It supports infrastructure development for **open repositories** and encourages the use of open-source software.
3. Italian Open Science Strategy within PNRR
- As part of the **National Recovery and Resilience Plan (PNRR)**, Italy invests in **digitalization and research infrastructure**, including platforms for open science and open-access repositories.
 - It funds projects that promote data sharing and interoperability.
 - For example, all the projects funded in the framework of the “Piano Nazionale di Ripresa e Resilienza (PNRR)” - Next Generation Europe Recovery Funds explicitly require and finance FAIR data management and open science activities. Overall these projects have received 4,3 billion of euro <https://www.osservatoriorecovery.it/i-decreti-di-assegnazioni-delle-risorse-pnrr-del-mur-missione-4-componente-2-investimenti-1-4-1-5-3-1/> and involve a large population of researchers and innovation actors. The mandates included in these projects largely contribute to raising awareness of open science and accelerating the transition towards this new approach.

2.4 In your country, is there a national implementation plan, strategy, policy or roadmap for the implementation of open science at the national level in line with the 2021 Recommendation?

(ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

YES / UNDER DEVELOPMENT / NO

YES

The National Plan for Open Science (Piano Nazionale per la Scienza Aperta - PNSA) was published in June 2022. It is available at the address

<https://www.mur.gov.it/it/news/lunedì-20062022/pubblicato-il-piano-nazionale-della-scienza-aperta>

https://www.mur.gov.it/sites/default/files/2023-01/PNSA_2021-27_ENG.pdf

PNSA is part of the National Plan for Research 2021-2027 developed by the Ministry of Research (MUR). It was prepared with the contribution of a group of selected experts from different disciplines and knowledgeable on different elements of open science.

PNSA highlights the need to implement synchronised actions along multiple dimensions:

- *General measures – i.e. the priority and preparatory actions to proceed with everything else;*
- *Regulatory indications – to support the implementation of the plan and the cultural and organizational change towards open science;*
- *Research results – for the effective and efficient sharing and management of the results of Italian research according to the FAIR principles;*
- *Research assessment – to support assessment methodologies and practices in line with an open approach to science;*
- *Infrastructure and services – to provide technological tools suitable for implementing all the plan elements. The set of tools is also expected to constitute the core of what could become the “National Infrastructure Node” of the EOSC federation for the years following 2027.*
- *Training – to create the skills necessary to achieve the objectives of the PNSA;*
- *European participation and the European Data Space – to connect to the European panorama and to the emerging initiatives that have the sharing of research products at their core;*
- *Research - to identify new solutions that natively respond to the need to make research work more effective and efficient as it will be conducted in the near future when open science will be the norm.*

In March 2023, an expert group was appointed by MUR with the task of preparing an implementation plan by 2027 in line with PNSA. The expert group was also requested to propose processes for identifying existing assets that could be effectively leveraged for its implementation.

[https://open-science.it/article?rpk=321298&prs_sel=p_researcher&tpc_sel=t_piano_nazionale_scienza_aperta]

The produced implementation plan involves a joint effort of, among others, academic and research institutions, the National Research Assessment Agency (ANVUR) and the Ministry of

University and Research. It also anticipated the contribution of specialised technical companies if needed.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at institutional level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d.17.e. 17.g)

YES /UNDER DEVELOPMENT/ NO

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or coproduction of knowledge with multiple actors).

YES

At least 75 academic and research institutions have adopted an Open Science policy. Currently, most of these policies focus on publications, with nine referencing FAIR Data and six addressing Data Management. By 2025, this number is expected to grow significantly.

All 75 policies, along with detailed descriptions, are available in a dedicated section of the [open-science.it](#) catalogue.

Additionally, some institutions have developed operational work plans to align internal activities and processes with these policies. Notable examples include the National Research Council (CNR): <https://zenodo.org/records/7983538>, <https://iris.cnr.it/handle/20.500.14243/510437>) and National Institute of Geology and Vulcanology (INGV) (<https://istituto.ingv.it/it/open-science.html>).

2.6 In your country, are there specific funding mechanisms for open science at the national level?

(ref.: (ii) 17.i. (iii) 18.i. (iv) 19. E /c)

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

YES. The Ministry of University and Research has allocated specific resources to support the implementation of the National Plan for Open Science.

2.7 In your country, have open science practices been integrated in existing research funding mechanisms?

YES/NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

As previously mentioned in response to question 2.3, an increasing number of national project funding programs now include mandates for FAIR data management and open science practices. The costs associated with these activities are considered eligible expenses.

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

YES / NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

*Yes, Italy is actively promoting equitable **public-private partnerships on open science** through investments in the **National Recovery and Resilience Plan (PNRR)**. By the end of 2025, the PNRR will have allocated significant resources to research and innovation, requiring compliance with **open science principles** and fostering collaboration between public institutions, universities, and private sector actors.*

*Specifically, the **Ministry of University and Research (MUR)** has designated **4.3 billion euros** to projects that integrate **FAIR data management** and **open-access policies**, ensuring that both public and private entities contribute to and benefit from open science practices. These projects are designed to **increase knowledge-sharing, enhance research accessibility**, and strengthen partnerships in line with the **2021 UNESCO Recommendation on Open Science**."*

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.i. 23.c)

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

UNDER DEVELOPMENT

There is no centralized national monitoring system for Open Science. However, several initiatives aim to track the adoption of Open Science practices in academic institutions and research-performing organizations. These initiatives have largely agreed on a shared set of core indicators while incorporating additional institution-specific metrics.

Monitoring efforts rely on open, non-proprietary, and transparent infrastructures provided by research institutions, such as those described at the following addresses:
https://services.d4science.org/web/osobservatory_it,
<https://www.openaccessrepository.it/record/143565>)

The monitored data are supplied and overseen by designated personnel within the participating institutions. When possible, additional data are gathered through the European OpenAIRE infrastructure (www.openaire.eu).

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

According to official data from the **National Institute of Statistics (ISTAT)**, Italy's expenditure on **Research and Development (R&D)** showed the following trends in recent years:

- **2020:** R&D expenditure represented **1.51% of GDP**, an increase compared to the previous year (1.46%). (istat.it)
- **2021:** A slight decrease was recorded, with R&D expenditure at **1.45% of GDP**. (istat.it)
- **2022:** R&D expenditure was **1.41% of GDP**, with a 5.0% increase in absolute terms compared to 2021. (istat.it)

These data highlight a growth in absolute terms for R&D expenditure in 2022, but a slight decrease in its percentage share of GDP compared to previous years.

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

According to official data, as of **2023**, Italy's **Fiber to the Home (FTTH)** coverage reached **59%** of the market, corresponding to approximately **15.5 million households**, with a penetration rate of **16%**. However, Italy ranks **25th in Europe** for FTTH coverage, and only **7.3%** of the territory is served by standalone 5G technology.

To address these challenges, the Italian government initiated the **Strategic Plan for Ultrabroadband** in 2016, aiming to extend ultrabroadband across the country. In **2020**, an agreement was signed between the **Ministry of Economic Development**, **Invitalia**, and **Infratel Italia** to enhance the public ultrabroadband infrastructure, focusing on under-served areas.

Furthermore, in **2023**, the **Interministerial Committee for Digital Transition** launched the **National Ultrabroadband Strategy**, planning investments up to **€2.8 billion** for the 2023-2026 period, aiming to accelerate the development of high-performance connectivity nationwide.

To monitor and update the ultrabroadband connectivity availability map, **Infratel Italia** periodically conducts public consultations with telecommunications operators, identifying areas with "market failure".

These initiatives highlight the Italian government's commitment to improving high-speed internet access, aiming to reduce the digital divide and ensure more equitable connectivity across the national territory.

3.3 Are there national research and education networks (NRENs) active in the country?
18.c)

YES / NO

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

YES

The National Research and Education Network (GARR) is designed and managed by Consortium GARR, a non-profit association operating under the auspices of the Ministry of University and Research. Its members include major research institutions such as CNR, ENEA, CRUI Foundation, INAF, INFN, INGV, Italian public universities, IRCCS (Scientific Institutes for Research, Hospitalization, and Healthcare), and IZS (Experimental Zooprophyllactic Institutes).

The GARR network interconnects universities, research centers, libraries, museums, schools, and other institutions dedicated to education, science, culture, and innovation across Italy, offering high-capacity connectivity.

This fiber-optic infrastructure, spanning approximately 20,000 km, integrates cutting-edge technologies through continuous experimentation and innovation. Due to its scalability, the network's backbone and access connections can seamlessly evolve alongside user needs.

GARR is also globally interconnected with research and educational networks, and as a founding member of GÉANT, it contributes to the European research fiber-optic backbone. GÉANT interconnects national research networks across Europe and provides high-capacity links to America, Asia, Africa, and Oceania.

3.4. Are there national or institutional open science infrastructures in your country?(ref.: (iii) 18. d.18.e. 18.k. 18.l, 18.m)

YES / UNDER DEVELOPMENT / NO

If yes or under development, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including highperformance computing, cloud computing and data storage.
- community managed infrastructures, protocols and standards, for example those that support biodiversity and engagement with society
- platforms for exchanges and co-creation of knowledge between scientists and society
- community-based monitoring and information systems to complement national, regional and global data and information systems.

For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial. If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

UNDER DEVELOPMENT

Academic and research-performing organizations are demonstrating a strong commitment to advancing open science by developing infrastructures that enhance data accessibility, foster collaborative research, and facilitate the dissemination of scientific knowledge.

Several research institutions have already established high-quality open science infrastructures, including:

- **Istituto Nazionale di Geofisica e Vulcanologia (INGV):** INGV actively promotes open science through accessible, interoperable, and sustainable infrastructures that support data sharing and interdisciplinary collaboration. Notable examples include its Open Science platform and the INGV Data Portal [<https://istituto.ingv.it/it/open-science.html> https://data.ingv.it/docs/index_en.html#document-doc/introduction]
- **Consiglio Nazionale delle Ricerche (CNR):** CNR has developed horizontal infrastructures to support open science, such as **D4Science** [www.d4science.org], which facilitates open science workflows, and **OpenAIRE** [www.openaire.eu], a European initiative dedicated to promoting open scholarly communication.
- **Istituto Nazionale di Fisica Nucleare (INFN):** INFN has developed **INFN Cloud** (cloud.infn.it), an open, extensible, federated cloud infrastructure designed to support a wide range of scientific research activities and facilitate open science practices. Additionally, INFN maintains an **Open Access Repository (OAR)** [<https://web.infn.it/openscience/cose-oar/>] to ensure broad access to research outputs. It is also largely involved in High Performance Computing Initiatives (e.g. ICSC at national (e.g. Centro Nazionale di Ricerca in HPC, Bigdata e Quantum Computing- ICSC <https://www.supercomputing-icsc.it/en/icsc-home/>) and international level (e.g. European High-Performance Computing Joint Undertaking - EuroHPC JU <https://eurohpc-ju.europa.eu/>))
- **Istituto Nazionale di Astrofisica (INAF):** INAF operates the **oa@INAF Open Access Repository** and maintains a **Cloud Science Platform** integrated into the **European Grid**

Infrastructure (EGI) Federated Cloud [www.egi.eu]. These infrastructures enable efficient data processing, storage, and sharing, further supporting open science initiatives.

At the national level, the National Plan for Open Science mandates the creation of a national Open Science infrastructure, developed collaboratively by academic institutions, research organizations, and industry partners. This initiative is led by ICSC (ICSC Supercomputing Center [<https://www.supercomputing-icsc.it/en/icsc-home/>]) and ICDI (Italian Computing and Data Infrastructure) [www.icdi.eu] and will leverage existing resources and institutional infrastructures.

Starting in 2025, the national infrastructure will be federated with other European nodes as part of the **European Open Science Cloud (EOSC) build-up phase** [<https://eosc.eu/news/2025/02/eoscs-build-up-phase-to-begin-with-march-kick-off/>], [https://research-and-innovation.ec.europa.eu/strategy/strategy-research-and-innovation/our-digital-future/open-science/european-open-science-cloud-eosc_en].

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 1!!!)

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

If no, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

YES

The Italian National Research Council(CNR) is a key member contributing to the development of the Open Access Infrastructure for Research in Europe (OpenAIRE - www.openaire.eu).It also hosts a set of IT services of this infrastructure.

Italy hosts services and funds a large number of European Research Infrastructures that are used by European and international researchers to perform their activities and produce their outcomes. All the Research Infrastructures are addressing to a different level of development aspects of Open Science.

Through initiatives like the Science-Clusters (<https://science-clusters.eu/>) these infrastructures are collaborating to identify common strategies and solutions to advance open science across Europe's scientific community.

Italy is also largely involved in the development and service provision of large European IT infrastructure serving research communities. Examples are Geant (<https://network.geant.org/>), ensuring network connectivity, to which Italy contribute through its national network Garr (<https://www.garr.it/it/infrastrutture/rete-nazionale/infrastruttura-di-rete-nazionale>) and EGI.eu whose national contribution comes from INFN (<https://www.egi.eu/partner/infn/>).

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science, stage of research, accessibility to all the relevant users in the region or internationally).

Yes, Italy is actively involved in collaborative initiatives to optimize infrastructure use and develop joint strategies for open science platforms at both national and international levels, involving various regions of the country.

D4Science is one of the main infrastructures, offering virtual research environments (VREs) to support open science practices. Managed by the CNR in Pisa, D4Science serves over 24,000 registered users and supports various scientific communities such as marine sciences, earth sciences, and agri-food.

ELIXIR Italy is a European infrastructure for sharing biological data, to which Italy actively contributes, facilitating collaboration between Italian scientific organizations for the management of biological data.

In addition to these initiatives, Italy promotes various collaborations between northern and southern regions, as well as between public and private entities, to improve access to and sharing of scientific data, supporting open science practices at the national level.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

YES / NO

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

One of the main collective initiatives for capacity building on open science at the national level is the Italian Computing and Data Infrastructure Competence Centre (ICDI-CC) [<https://www.icdi.it/en/activities/competence-centre>]. It gathers together stakeholders and practitioners belonging to twenty-one of the main Italian universities, research centres and infrastructures. It serves as a collaborative network aimed at supporting Italy's national

community in Open Science, the application of FAIR (Findable, Accessible, Interoperable, Reusable) principles, and participation in the European Open Science Cloud (EOSC) initiative.

Its activities include:

- **Expert Network:** ICDI-CC brings together professionals with diverse competencies to assist stakeholders in adopting Open Science practices, implementing FAIR principles, and engaging with EOSC (<https://zenodo.org/records/5512638>)
- **Support Services:** The Centre offers services and tools tailored to the emerging needs of the community, including a catalogue of resources and courses [<https://publications.cnr.it/doc/481051>]
- **Stakeholder Engagement:** ICDI-CC addresses a variety of stakeholders, such as governance bodies of research institutions, funding agencies, individual researchers, support staff, public administrations, enterprises, and citizens, promoting the benefits of Open Science and FAIR

The ICDI-CC has given birth, among the others, to the Italian Data Stewards network that today includes more than 100 participants https://open-science.it/it/article?rpk=313614&prs_sel=p_datasteward&tpc_sel=t_gestione_dei_dati_della_ricerca

ICDI-CC is part of a broader European network of Competence Centres that collaborate to advance Open Science and EOSC initiatives. It participates in the Skills4EOSC EU project [skills4eosc.eu], which aims to create a training ecosystem for Open and FAIR Science by connecting national Competence Centres across Europe. This collaboration facilitates the sharing of best practices, resources, and strategies to enhance the implementation of Open Science policies and infrastructures continent-wide.

Through these efforts, ICDI-CC contributes to the development of a cohesive European Open Science environment, fostering interoperability and collaboration among various national and institutional initiatives [<https://www.icdi.it/en/activities/competence-centre>].

In addition to the activities organised by the ICDI-CC competence centers several Italian universities offer courses and training programs focused on open science and FAIR (Findable, Accessible, Interoperable, and Reusable) data management. Notable examples include:

- **University of Milan (Università degli Studi di Milano)** [rdm.unimi.it]

The University provides training webinars on research data management aligned with FAIR principles. These sessions cover topics such as data management planning, metadata standards, and the use of institutional repositories like Data@UNIMI for data sharing.

- **University of Bologna (Università di Bologna)** [<https://www.unibo.it/it/ricerca/open-science>]:

The University promotes open science through initiatives that emphasize transparency, inclusion, research integrity, and collaboration. While specific course details are not provided, the institution's commitment to open science suggests the integration of these principles into its academic offerings.

- **International School for Advanced Studies (SISSA)** [<https://www.sissa.it/training/professional-master-courses/master-data-management-and-curation/>]:

SISSA offers a Specialization in Data Management and Curation (MDMC), focusing on open science principles, FAIR data management, and tools for data acquisition and metadata enrichment.

- **University of Turin (Università degli Studi di Torino)** <https://hpc4ai.unito.it/cuap-data-steward/>

The university provides courses on FAIR data management, emphasizing the importance of good data practices in research. They include practical exercises on drafting data management plans for real datasets. It also offers a first example of a course to upskill data stewards.

- **Scuola Normale Superiore (SNS)** <https://www.sns.it/en/transversal-activities-shared-all-phd-course>

SNS offers courses on Open Science and FAIR data management. Starting from the academic year 2022/23, the course "Open Science and Research Data Management (and use of IRIS)" has been mandatory for second-year Ph.D. students. This course is part of the transversal activities shared across all Ph.D. programs at SNS.

The above-mentioned initiatives are at the moment not fully coordinated. However, they demonstrate a growing commitment among Italian universities to equip researchers with the necessary skills and knowledge to manage research data responsibly and in alignment with open science and FAIR principles.

4.2 Have capacity building programmes for policy makers on how to integrate core values and

principles of open science in STI policies and strategies taken place in your country?

YES / NO

NO

There is no systematic programme for capacity building. However, the topic has been addressed in events (e.g. <https://sites.units.it/transform4europe/index.php/2024/05/14/exploring->

mechanisms-and-best-practices-for-fostering-public-engagement-with-citizen-science-science-communication-and-open-science/)

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025?

(ref.: (iv) 19.b)

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

NO

In Italy, the integration of Open Science competencies into higher education curricula is progressing, though a standardized national framework has yet to be universally adopted. The Italian National Plan for Open Science 2021-2027 sets objectives and recommendations on open access, research data management, and evaluation practices to promote Open Science principles.

Despite these strategic initiatives, a comprehensive framework for embedding Open Science competencies across all higher education institutions in Italy is still under development. Individual universities and research organizations are increasingly incorporating Open Science and FAIR data management training into their programs, but the extent and structure of these offerings vary. Continued efforts are underway to harmonize these competencies within the national higher education curriculum.

4.4 Are open educational resources as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

YES/ PARTLY/ NO

NO

Despite Italian experts take part in international initiatives, like Skill4EOSC [<https://www.skills4eosc.eu/>], promoting OER, no systematic approach is still in place.

4.5 Have there been any initiatives in your country to support science communication in line with

open science value and principles with a view to the dissemination of scientific knowledge to

researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

There are initiatives and groups of interest that are discussing and taking action in this direction. Examples of these initiatives are:

- **Associazione Italiana per la Promozione della Scienza Aperta (AISA)** [<https://aisa.sp.unipi.it/chi-siamo/>]: AISA is dedicated to disseminating open science culture in Italy. Its mission includes publishing studies on open science implementation, providing training programs for research organizations, and engaging with policymakers to promote open science in research assessment and intellectual property policies.
- **Citizen Science Italia ETS** [<https://www.citizenscience.it/en/>]: This national network connects initiatives, experts, universities, research centers, museums, associations, and public bodies to build inclusive and collaborative tools in line with open science perspectives. It aims to foster citizen engagement in scientific research, enhancing transparency and public participation.

The importance of responsible science communication is highlighted in the G7 Science and Technology Ministers' Meeting Communiqué adopted last July in Bologna, under the Italian G7 Presidency. [<https://www.g7italy.it/wp-content/uploads/G7-Science-and-Technology-Ministers-Meeting-Communiqué.pdf>]. The communiqué underlines the role of scientific communication in disseminating knowledge and emphasizes the need for dialogue and relationship-building between the scientific community and the public. Strengthening this connection fosters public trust and supports the development of well-informed, evidence-based policies with greater societal impact.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20c.)

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so

YES

Italy is undertaking several initiatives at both national and institutional levels to align research assessment and evaluation processes with open science principles and values

The review of existing research assessment and evaluation is included in the National Plan for Open Science.

Fifty-three between Italian academic and research performing institutions participate in the International Coalition for Advancing Research Assessment (CoARA) [<https://coara.eu/>]. Together they have established the CoARA Italian Chapter [<https://www.coara-italia.it/>]. It supports the implementation of research assessment reforms at the national level and it aims to attract more institutions and stakeholders to endorse CoARA's agreement. Its mission includes the following main actions: [https://coara.eu/app/uploads/2023/10/NC-Overview_Italy_06Oct23.pdf?utm_source=chatgpt.com]:

- 1. **Enable Mutual Learning and Share Best Practices:** Facilitate the exchange of knowledge and raise awareness about responsible assessment practices within the national research community.*
- 2. **Foster Discussion on Assessment Criteria Development:** Encourage dialogue on reviewing and developing assessment criteria, tools, and processes for evaluating research institutions, individual researchers, and projects.*

Steps towards implementing a review process of the assessment practices have been made by a number of research institutions. The changes reflect an effort to reform research assessment and evaluation processes, ensuring they align with open science principles and values, thereby promoting transparency, inclusivity, and collaboration in scientific research. Examples of institutions where the revision is in progress are:

- **University of Bologna** [https://www.unibo.it/en/research/research-evaluation?utm_source=chatgpt.com] Key points of its Action Plan Key include the promotion of open science, the valorization of a wider range of research-related contributions, the importance of qualitative evaluation, and the need to renew tools, criteria, and processes associated with research evaluation.*
- **University of Turin** [https://en.unito.it/research/research-policies-and-facilities/open-science?utm_source=chatgpt.com]: The University of Turin emphasizes that adopting open science requires reviewing traditional research methodologies, shifting from a focus on publishing to sharing all research elements as early and openly as possible, including co-creation of knowledge in dialogue with society.*

- **National Research Council (CNR)** [<https://publications.cnr.it/doc/482044>] : The CNR Roadmap for Open Science outlines a path for implementing open science practices and policies within the institution. The Roadmap includes a plan to define a rich, varied and constantly updated set of quality and impact indicators to evaluate researchers, projects, groups/institutes and the Institution itself, taking into account the mission and the various institutional activities.

The Italian National Agency for the Evaluation of Universities and Research Institutes (ANVUR) has signed the CoARA Agreement, signaling a commitment to better recognize diverse research practices and to move away from inappropriate uses of metrics. Very preliminary steps towards the changes suggested by CoARA have been implemented in the published guidelines for the next evaluation round concerning the period [2020-2024] [https://unikore.it/wp-content/uploads/2024/12/Bando-VQR-2020-2024_31ottobre.pdf].

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a. 20.c. 20.d)

YES / NO

If yes, please provide details and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

NO

Only preliminary steps have been taken in an uncoordinated and experimental manner by a few academic and research institutions, such as CNR, as part of its Roadmap for Open Science [<https://open-science.it/-/il-cnr-approva-la-roadmap-per-la-scienza-aperta>, <https://zenodo.org/records/7983538#.ZHcC9OxBw6E>].

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a. 20.b, 20.c, 20.e. 20.f. 20.g, 20h)

YES / NO

If yes, for each initiative, please provide more information, including on the stakeholders that are involved or lead the initiative, and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

Fifty-three Italian academic and research-performing institutions are members of the International Coalition for Advancing Research Assessment (CoARA) [<https://coara.eu/>], demonstrating their commitment to implementing reforms that recognize and reward open science practices.

5.4 Have there been any unintended negative consequences of open science practices reported in your country? (ref.: (v) 20)

YES / NO

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

YES

In Italy, there is an ongoing debate on balancing open science with security and research integrity. A dedicated workshop was organized in 2024 to explore this issue in preparation for the G7 Science and Technology Ministers' Meeting in July 2025 [https://www.mur.gov.it/sites/default/files/2024-03/Castelli_Workshop_SIGRE_2023.11.29.pdf]. The topic is also reflected in the outcomes of this Ministerial meeting [<https://www.g7italy.it/wp-content/uploads/G7-Science-and-Technology-Ministers-Meeting-Communique.pdf>].

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

YES, at the national level.

YES, at the institutional level.

YES, at both national and institutional levels.

NO

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant:

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.
- promoting open innovation practices.
- Others (with a box to add text)

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

Several initiatives at the national and institutional levels are driving changes across different stages of the research cycle to enhance the openness of the entire scientific process. Some key examples include:

- **Italian Reproducibility Network (ITRN)** [<https://www.itrn.org/>] – As the Italian branch of the international reproducibility network [<https://osf.io/aq5je>], ITRN is dedicated to improving research quality and fostering open science practices in Italy. Among its activities, ITRN conducts and publishes surveys and empirical studies on open science implementation, providing insights into current practices and identifying areas for improvement. (promoting open innovation practices, valuing and sharing of negative scientific results and associated data, valuing and sharing of negative scientific results and associated data)
- **Citizen Science Italia ETS** [<https://eu-citizen.science/organisation/302>] This national network aims to advance citizen science in Italy while actively integrating open science principles into its mission and activities [<https://www.youtube.com/watch?v=WHodib58qng>]. Comprising initiatives, experts, universities, research centers, scientific museums, associations, and public bodies, the association works to develop inclusive and collaborative tools and solutions within an open science framework. (promoting open innovation practices, participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific project)
- **OpenScience@CNR** – The National Research Council of Italy (CNR) has secured public funding to implement its Roadmap for Open Science [<https://open-science.it/-/il-cnr-approva-la-roadmap-per-la-scienza-aperta>]. The **OpenScience@CNR** project is particularly focused on developing new methods, tools, and practices to support researchers in adopting open science approaches. (promoting open innovation practices)

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

YES

International collaboration is a key action area in the National Plan for Open Science and is also reflected in its implementation workplan. This collaboration is encouraged through incentives for participation in international initiatives on open science, significant contributions to the development and operation of EU and global research infrastructures, and active engagement in the European Open Science Cloud.

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e. 22.f)

YES / NO

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

YES

The National Plan for Open Science recognizes the importance of cross-border, multi-stakeholder collaboration in advancing open science. Several initiatives are already underway to support this objective. Below are some key examples:

- **Italy's Role in the European Open Science Cloud (EOSC)**
*Italy is actively contributing to the build-up phase of the **European Open Science Cloud (EOSC)** ((<https://eosc.eu/eosc-about/building-the-eosc-federation/contributing-to-the-build-up-phase-of-the-eosc-federation/>).). Among the thirteen organizations selected as part of the first wave of nodes launching the EOSC Federation, two are Italian:*
 - **CNR-Blue-Cloud 2026** ((https://www.blue-cloud.org/news/cnr-blue-cloud-2026-selected-eosc-federation-build-phase)))
 - **Foundation ICSC** (<https://www.supercomputing-icsc.it/en/2024/12/06/icsc-italian-eosc-node-selected-to-join-the-eosc-federation-as-potential-candidate-node/>)
These organizations will play a crucial role in shaping the EOSC infrastructure by providing data, services, and computing resources for open science at the European level.
- **Italy's Commitment to Open Research Europe (ORE)**
*The **Italian Ministry of Research (MUR)** has announced its intention to join a coalition of national research funding and performing organizations from eight European countries in supporting **Open Research Europe (ORE)**—a collective, not-for-profit open-access publishing platform. Italy's participation aims to ensure sustainable funding for ORE for an initial five-year period, starting in 2026 (https://open-science.it/article?rpk=333456&prs_sel=p_institution&tpc_sel=t_openscience).*

Through these and other initiatives, Italy is strengthening its engagement in international open science efforts, fostering collaboration across borders, and ensuring broader access to research outputs.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.:

YES / NO

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Yes, Italy is actively involved in discussions on the establishment of regional and international funding mechanisms for open science. A key example is Italy's participation in **Plan S**, which promotes full and immediate open access to scientific publications, with the **National Institute for Nuclear Physics (INFN)** among its members. Additionally, Italy is part of the EU's **COST** program, which funds scientific collaboration and supports open science practices. The **National Research Council (CNR)** also supports scientific and technological research in collaboration with international institutions, contributing to the global open science initiative.

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES/NO

If yes, please provide more information.

Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, opensource software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

No, there are no specific external causes impeding the implementation of the 2021 UNESCO Recommendation on Open Science in Italy. However, there is growing attention to the ethical aspects of Artificial Intelligence and Cybersecurity. The solutions introduced to address the various elements of open science must be compatible and aligned with those for these two areas. A multidisciplinary approach, along with dedicated research programs, is needed to identify solutions that can meet the requirements across these different domains. This ensures that open science initiatives can be integrated effectively while maintaining the necessary ethical and security standards.

